

Smart Candidate Screening System

Project Report

Purpose: Transparent, consistent, and explainable student screening for placement teams.

Technology: Python, Pandas, Streamlit, and CSV data.

Live application: <https://gamepatt78-smart-candidate-screening-system-app-1fibjl.streamlit.app/>

GitHub repository: <https://github.com/gamepatt78/-smart-candidate-screening-system>

Local preview: <http://127.0.0.1:8501>

1. Introduction

The Smart Candidate Screening System is a transparent web application for helping placement coordinators review student profiles quickly and consistently. It replaces slow, manual spreadsheet screening with a shared rule-based scoring process. Each candidate is evaluated using the same written criteria, and the application explains the reason for every category assigned.

The system supports human decision-making rather than replacing it. Coordinators can filter candidates by identity, branch, CGPA, skills, and readiness category before preparing a recruitment shortlist.

2. Problem Statement

Manual candidate screening becomes unreliable as the number of students increases. Two coordinators reviewing the same dataset may reach different conclusions because the meaning of a “strong” candidate is not formally defined. Strong students can be overlooked when reviews are rushed or inconsistent, while recruitment drives often require a shortlist before there is enough time to thoroughly review every student.

Need: A fast, consistent, explainable screening tool that applies shared rules and lets coordinators narrow candidates without manually scanning a spreadsheet.

3. Project Objectives

1. Create a shared definition of candidate readiness.
2. Apply the same evaluation rules to every student.

3. Provide a transparent explanation for each result.
4. Reduce the time required to prepare recruitment shortlists.
5. Allow coordinators to locate candidates using practical filters.
6. Present candidate information in a clear dashboard.

4. Proposed Solution

The application reads candidate information from a CSV file, cleans and standardizes the data, calculates a score for each student, and displays the results in an interactive Streamlit dashboard.

It uses deterministic rules rather than a black-box artificial intelligence model. The same candidate data always produces the same score and category, allowing a coordinator to inspect and explain the result in one sentence.

5. Evaluation Method

Each candidate can receive a maximum of 9 points:

Evaluation area	Scoring rule	Maximum
CGPA	2 points for 8.5 or above; 1 point for 7.0 to 8.49	2
Skills	2 points for 4 or more skills; 1 point for 2 to 3 skills	2
Projects	2 points for 3 or more projects; 1 point for 1 to 2 projects	2
Internship	2 points when internship experience is listed	2
Certifications	1 point when at least one certification is listed	1
Total		9

- **Strong:** 7 to 9 points
- **Average:** 4 to 6 points
- **Needs Improvement:** 0 to 3 points

6. Main Features

Candidate dashboard

Displays the candidate list, total students shown, and counts for each readiness category.

Search

Searches by candidate name, ID, or branch. Numeric ID searches use exact matching, so searching for ID 2 does not return ID 12.

CGPA range filtering

The range slider supports minimum and maximum values, such as 3.40 to 6.60. The reset button restores the visible range to 0.0 to 10.0.

Skills and category filtering

Coordinators can find candidates with skills such as React and combine this with CGPA and category filters.

Explainable student details

Each candidate has expandable details containing contact information, branch, CGPA, skills, projects, internship experience, certifications, and a scoring explanation.

7. Technical Implementation

- **Python:** application logic
- **Streamlit:** interactive web interface
- **Pandas:** CSV loading, cleaning, filtering, and calculations
- **CSV:** candidate dataset storage

The system cleans missing values, converts CGPA values into numbers, separates list fields such as skills and projects, calculates scores, assigns categories, and applies the selected filters.

8. Benefits

- Faster shortlist preparation
- More consistent evaluation across coordinators
- Clear and defensible placement decisions
- Less manual spreadsheet scanning
- Easy combination of academic and technical filters
- Better visibility into each student profile

9. Limitations

The system is a screening aid, not a complete recruitment decision-maker. Its results depend on accurate and complete CSV data. The simple scoring rules may not capture communication ability, interview performance, leadership, or role-specific requirements. Authentication, direct CSV upload, shortlist export, recruiter notes, and role-specific scoring are future enhancements.

10. Future Enhancements

- Upload and validate new candidate files
- Export filtered shortlists to CSV or PDF

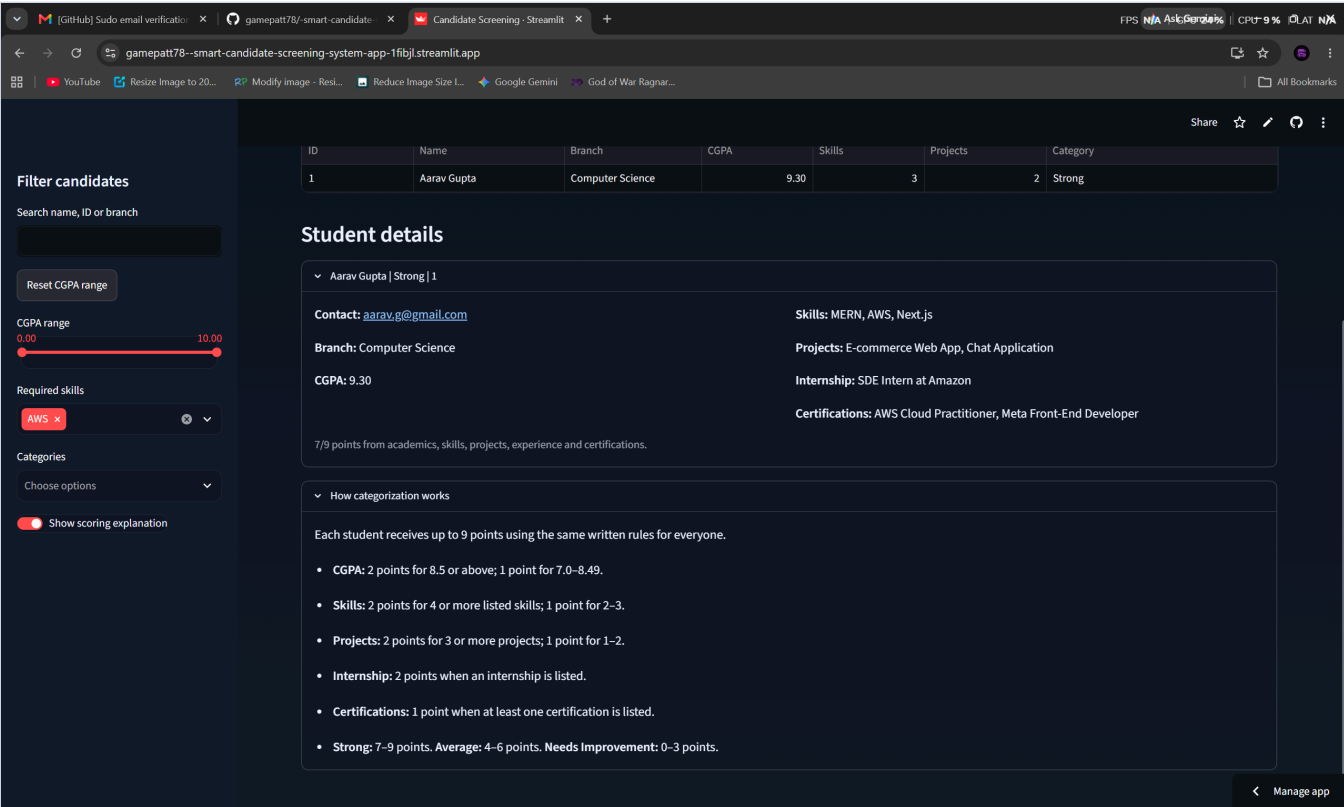
- Role-specific rules for different recruitment tracks
- Recruiter notes and shortlist status tracking
- Authentication and coordinator-level access control
- Audit history for scoring rule changes

11. Conclusion

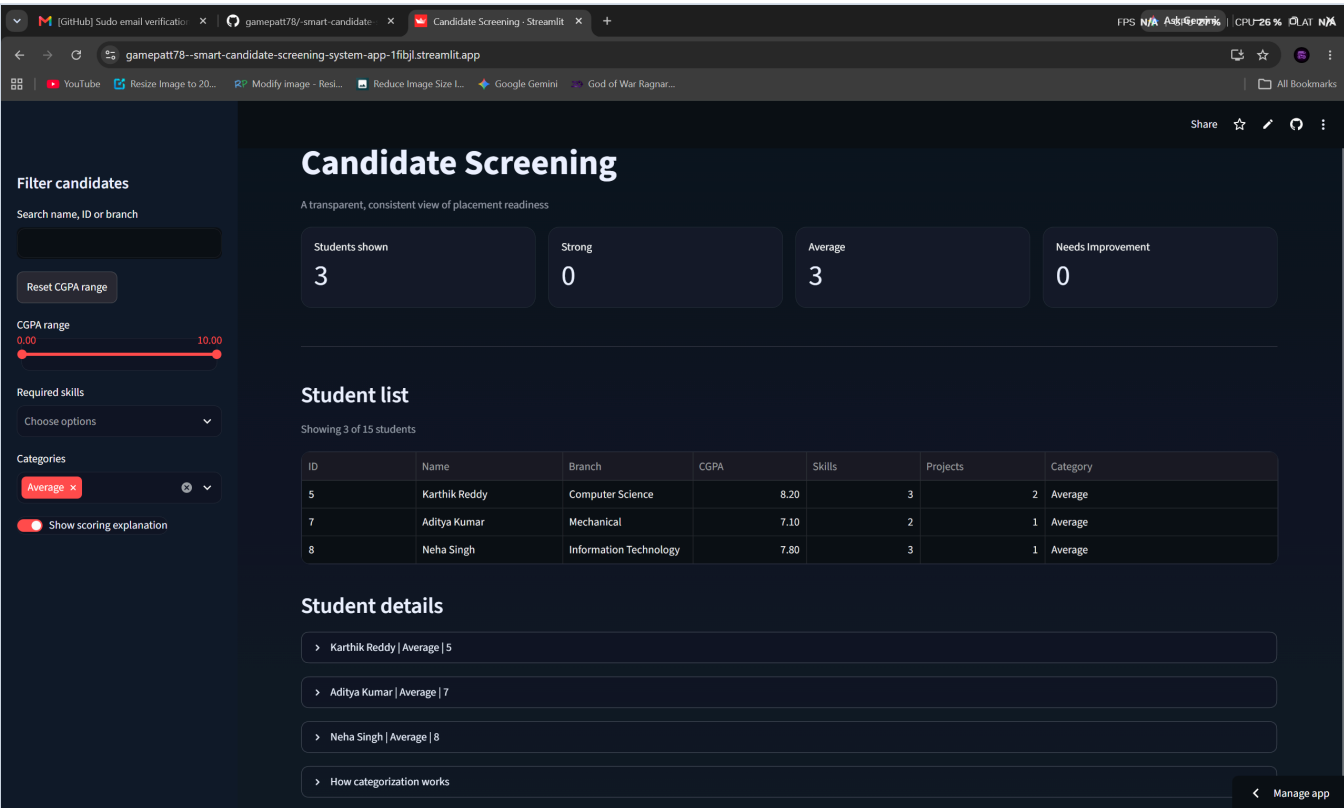
The Smart Candidate Screening System addresses inconsistency, delay, and lack of explainability in manual candidate review. By applying written scoring rules to every student and providing practical filters, it helps placement coordinators prepare fair shortlists more efficiently.

The system does not make mysterious decisions. It provides a transparent starting point that coordinators can inspect, explain, and use alongside professional judgment.

12. Screenshots and Product Views



Dashboard overview with summary metrics and candidate filters.



Expandable student details and scoring explanation.

Filter candidates

Search name, ID or branch

Reset CGPA range

CGPA range

Required skills

Categories

Show scoring explanation

Candidate Screening

A transparent, consistent view of placement readiness

Students shown

1

Strong

0

Average

0

Needs Improvement

1

Student list

Showing 1 of 15 students

ID	Name	Branch	CGPA	Skills	Projects	Category
14	Meera Reddy	Chemical	6.30	1	0	Needs Improvement

Student details

> Meera Reddy | Needs Improvement | 14

> How categorization works

Manage app

Filtered candidate results.

Filter candidates

Search name, ID or branch

Reset CGPA range

CGPA range

Required skills

Categories

Show scoring explanation

Candidate Screening

A transparent, consistent view of placement readiness

Students shown

6

Strong

0

Average

0

Needs Improvement

6

Student list

Showing 6 of 15 students

ID	Name	Branch	CGPA	Skills	Projects	Category
9	Vivek Joshi	Computer Science	6.90	2	1	Needs Improvement
11	Amit Chawla	Civil	5.90	1	0	Needs Improvement
12	Suman Rao	Mechanical	6.10	2	1	Needs Improvement
13	Deepak Tiwari	Electronics	5.40	1	0	Needs Improvement
14	Meera Reddy	Chemical	6.30	1	0	Needs Improvement
15	Rajeev Menon	Mechanical	5.20	1	0	Needs Improvement

Student details

> Vivek Joshi | Needs Improvement | 9

> Amit Chawla | Needs Improvement | 11

Manage app

Transparent scoring rules shown inside the application.

Filter candidates

Search name, ID or branch

Reset CGPA range

CGPA range

0.0010.00

Required skills

Choose options

Categories

Choose options

Show scoring explanation

Candidate Screening

A transparent, consistent view of placement readiness

Students shown

15

Strong

4

Average

3

Needs Improvement

8

Student list

Showing 15 of 15 students

ID	Name	Branch	CGPA	Skills	Projects	Category
6	Sneha Desai	Electronics	7.60	3	1	Needs Improvement
7	Aditya Kumar	Mechanical	7.10	2	1	Average
8	Neha Singh	Information Technology	7.80	3	1	Average
9	Vivek Joshi	Computer Science	6.90	2	1	Needs Improvement
10	Pooja Patel	Civil	7.40	2	1	Needs Improvement
11	Amit Chawla	Civil	5.90	1	0	Needs Improvement
12	Suman Rao	Mechanical	6.10	2	1	Needs Improvement
13	Deepak Tiwari	Electronics	5.40	1	0	Needs Improvement
14	Meera Reddy	Chemical	6.30	1	0	Needs Improvement
15	Rajeev Menon	Mechanical	5.20	1	0	Needs Improvement

Student details

Manage app

Candidate list with CGPA, skills, projects, and category.